

ME 630: Machine Learning for Engineers

Course Code	ME 630
Course Name	Machine Learning for Engineers
L-T-P-C	2-2-0-3
Prerequisites	None
Intended for	B.Tech/MTech/ MTech(R)/MS(R)/PhD
Mutual Exclusion	None
Approval	57th BoA (Oct 2024)

Course Contents

- Introduction: Various learning paradigms, definitions, examples. [1hr]
- Data handling: Data cleaning, Visualization and Pre-processing, Data Normalization, Linear Regression, Polynomial Regression [3hr]
- Optimization: gradient-based techniques, metaheuristic techniques, numerical implementation [4hrs]
- Statistics and probability: Probability distributions, hypotheses testing, Bayes' theorem, Maximum likelihood parameter estimation, Bayesian parameter estimation [3hrs]
- Supervised learning: Linear/nonlinear regression, overfitting, regularization, logistic regression, naive Bayes, k-NN, decision tree, random forest, maximum likelihood, support vector machine, applications in engineering [7hrs]
- Unsupervised learning: Singular value decomposition, principal component analysis, clustering, applications in engineering [5hrs]
- Artificial neural network: Single- and multi-layer networks, activation, backpropagation, stochastic gradient descent, physics-informed neural network, applications in engineering [5hrs]

List of Lab Experimentation:

It is suggested that the student should get hands-on experience by solving data analysis problems available on Machine Learning competition platforms. Some of the suggestive list of lab experiments is given below.

- Introduction to Python/MATLAB
- Extract the data from database
- Implement Support Vector Machine for a dataset and compare the accuracy by applying the following kernel functions:
i. Linear ii. Polynomial iii. RBF
- Implement k-nearest neighbours classification
- Implement linear regression using
- Implement K-Means_Clustering using
- Implement Naive Bayes Theorem to Classify the English Text
- Implement an algorithm to demonstrate Back Propagation Algorithm
- Implement an algorithm to demonstrate the significance of Genetic Algorithm in python
- Implement an algorithm to demonstrate Back Propagation Algorithm
- Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples
- Project work

Text Books

1. R. G. McClarren, **Machine Learning for Engineers**, Springer
2. S. Røgera, M. Girolami, **A First Course in Machine Learning**, CRC Press

References

1. Z-H. Zhou, **Machine Learning**, Springer
2. G. James et al., **An Introduction to Statistical Learning**, Springer
3. S. L. Brunton, J. L. Kutz, **Data-Driven Science and Engineering**, Cambridge Uni. press
4. J-A. Goulet, **Probabilistic Machine Learning for Civil Engineers**, MIT Press
5. J. Watt et al., **Machine Learning Refined**, 2nd Edition, Cambridge University press
6. A. Lindholm et al., **Machine Learning**, Cambridge University press